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ΔG = ?

$\Delta G = \Delta H - T\Delta S$

ΔG = – 10 kJ mol^{-1} – 300 K × $(-20x10^{-3})$ kJ K^{-1} mol^{-1}

ΔG = (–10+6) kJ mol^{-1}

ΔG = – 4 kJ mol^{-1}

At 600 K

ΔG = – 10 kJ mol^{-1} – 600 K × $(-20x10^{-3})$ kJ K^{-1} mol^{-1}

ΔG = (–10 + 12) kJ mol^{-1}

ΔG = + 2 kJ mol^{-1}

The value of ΔG is negative at 300K and the reaction is spontaneous, but at 600K the value ΔG becomes positive and the reaction is non spontaneous.

EVALUATION

Evaluation:

I Choose the best answer:

1. (b) ΔH
2. (d) decrease in free energy
3. (b) q = 0
4. (d) = 0
5. (a) $w = -\Delta U$
6. (d) $\frac{\text{mass}}{\text{volume}}$
7. (a) – 900 J
8. (b) negative
9. (b) – 67.6 kcal
10. (a) graphite is more stable than diamond
11. (d) – 462 kJ
12. (d) frictional energy
13. (d) $\Delta H < \Delta U$
14. (c) + 3 kJ
15. (a) – 2.48 kJ
16. (b) – 500 R
17. (d) $\frac{b-2a}{2}$
18. (d) – 635.66 kJ
19. (c) 80 kJ mol^{-1}
20. (a) $\Delta H < 0$ and $\Delta S > 0$
21. (c) adiabatic expansion
22. (d) (–, –, +)
23. (b) 27 °C
24. (d) $CaCO_3(S) \rightarrow CaO(S) + CO_2(g)$
25. (a) 300K